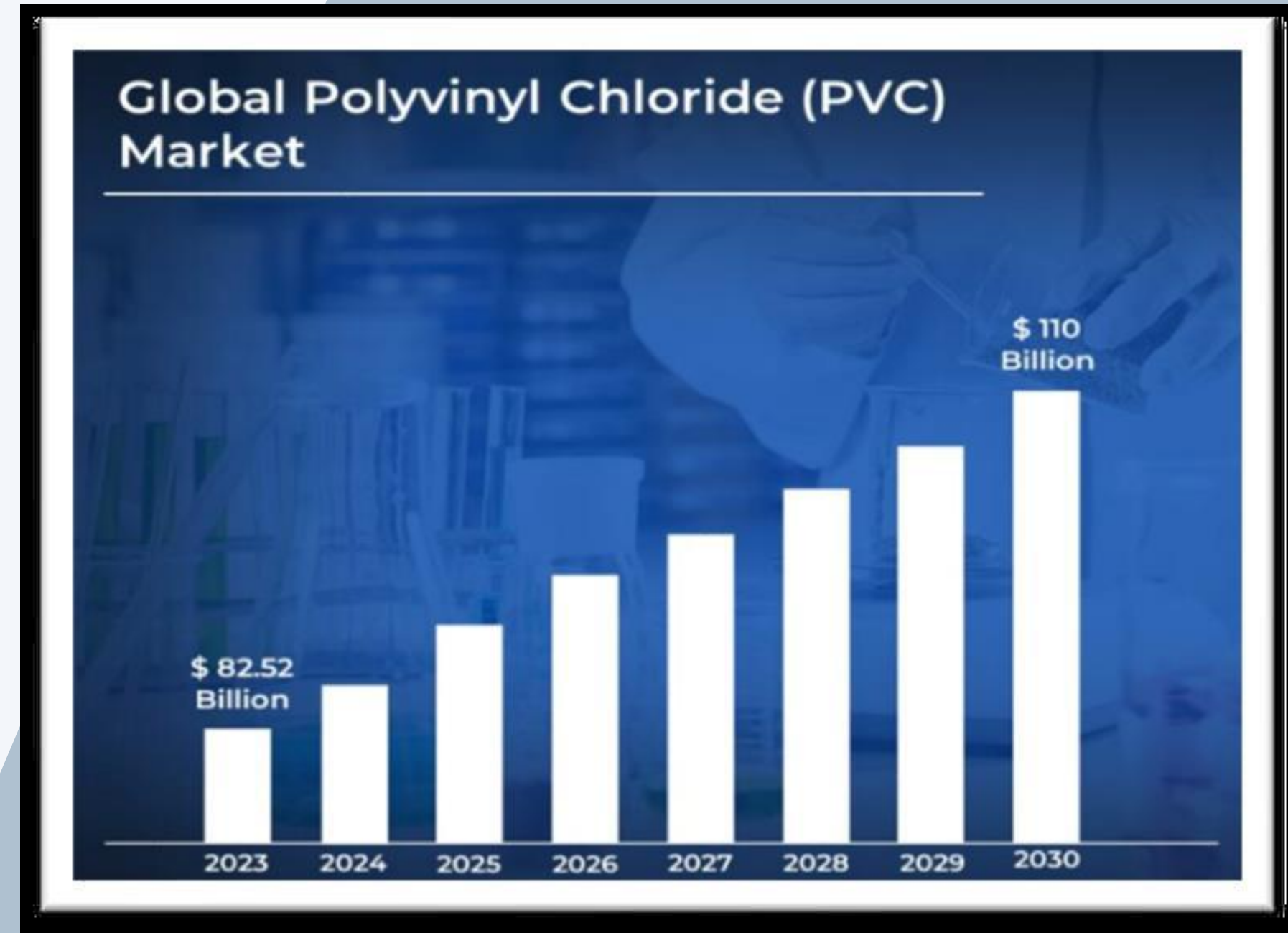


OPTIMIZING UV STABILIZATION IN PVC USING HALS



Introduction

- Polyvinyl Chloride (PVC) is a durable, cost-effective thermoplastic polymer widely used across industries due to its chemical resistance and versatile applications.
- It is prone to UV-induced degradation, which can compromise its performance, particularly in outdoor environments.





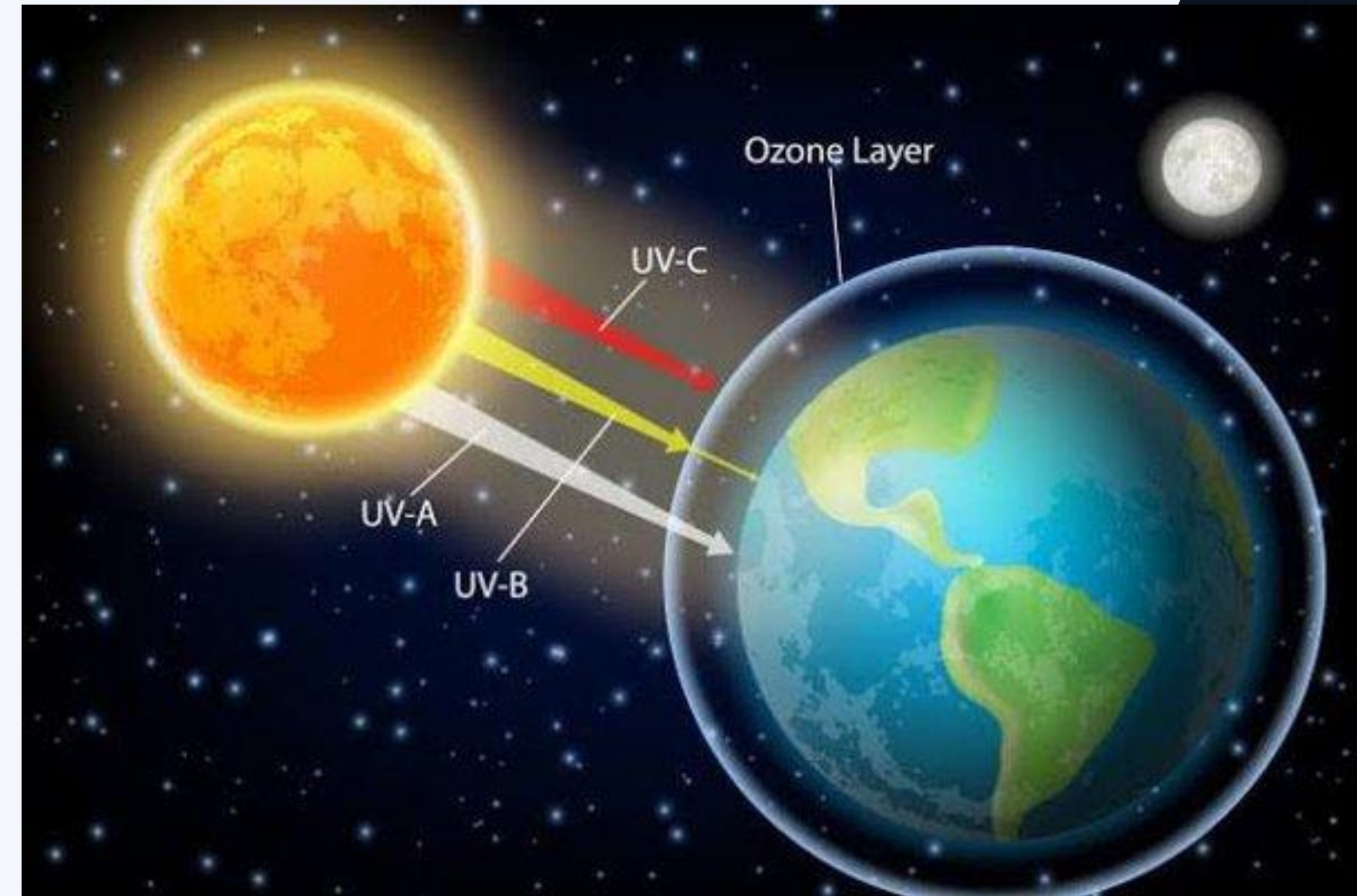
Objectives



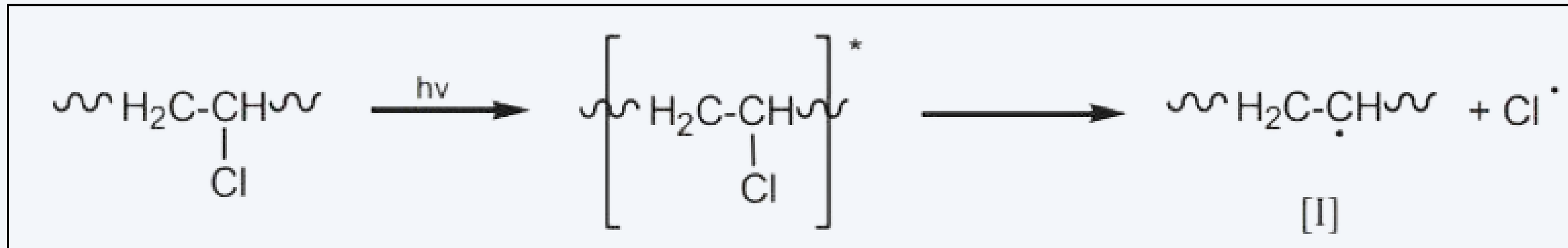
- Developing a kinetic model for PVC degradation under UV exposure.
- Integrating the action of HALS (Tinuvin 770) into the model.
- Creating a tool for predicting the optimal stabilizer concentration.

Mechanisms of Photodegradation

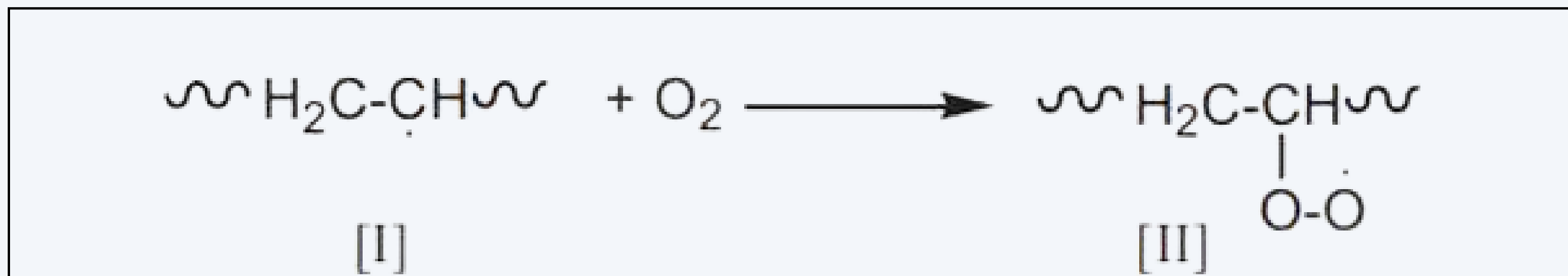
- UV Absorption
- Free Radical Formation
- Chain Scission & Cross-Linking
- Photooxidation



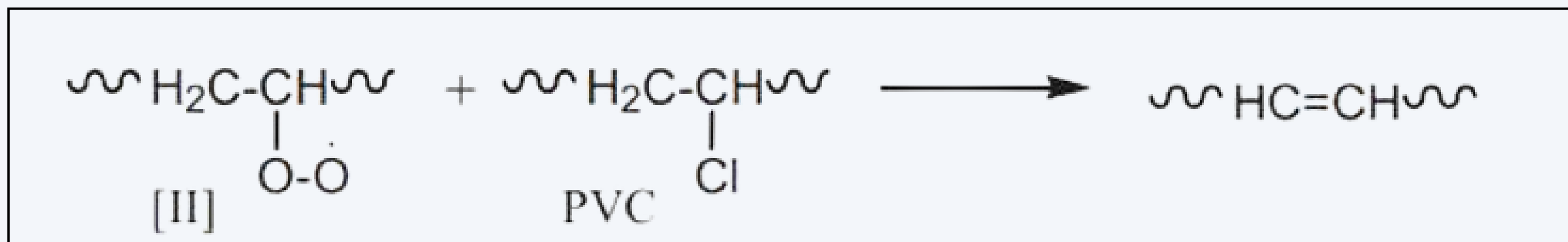
Mechanistic Pathways in UV Degradation



Formation of Polyene Radicals



Formation of Peroxy Radicals



Reaction of Peroxy Radical with PVC

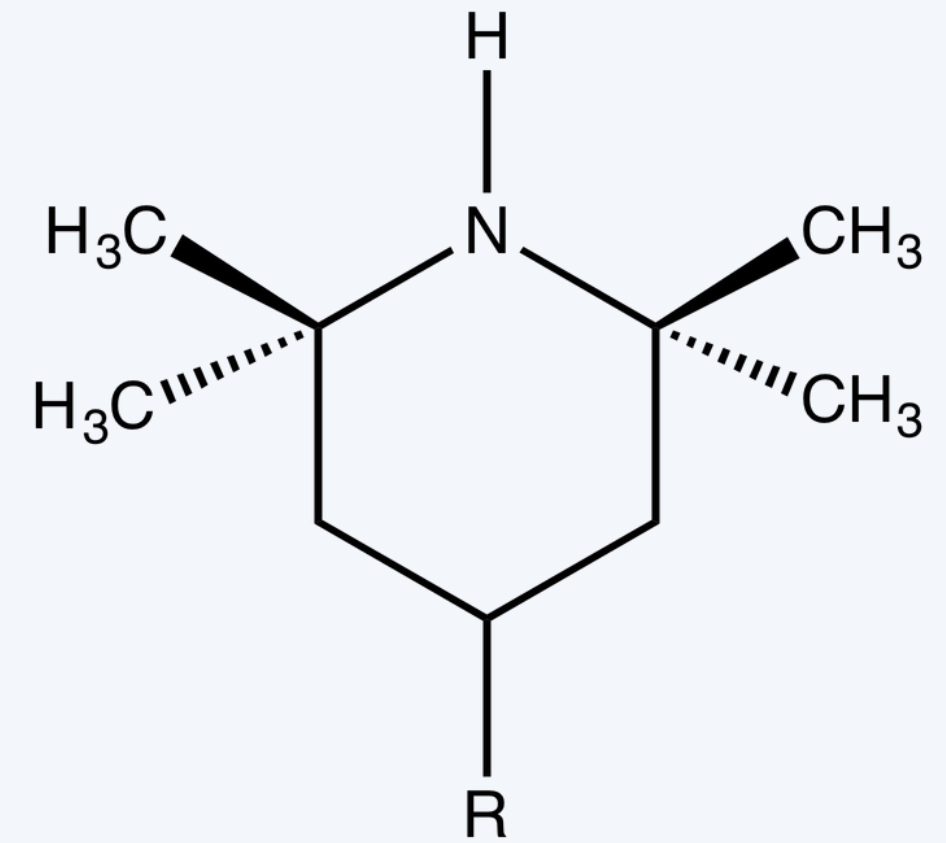
Effects of UV Degradation on PVC Properties

- Discoloration
 - Yellowing
 - Surface Cracking
- Loss Of Mechanical Properties
- Surface Roughness



Role of Stabilizers

- UV absorbers - absorbs UV rays and convert that energy to harmless amount of heat
- HALS - trap and neutralize the radicals
 - control the propagation process



Structure of HALS

Experimental Data On PVC Degradation

Accelerated Weathering Tests

UV Exposure Chambers

- Simulate real-world conditions (UV radiation, temperature, humidity) to predict long-term performance.

Spectroscopic Analysis

FTIR Spectroscopy

- Detects chemical changes, such as oxidation, by measuring absorption spectra.

UV-Vis Spectroscopy

- Assesses UV absorber effectiveness by measuring light absorbance.



Kinetic Analysis of Photodegradation

Rate equations without stabilizers,

Concentration of PVC:

$$\frac{d[PVC]}{dt} = -k_6 [PVC] - k_8 [PVC] [R'OO \bullet] - k_9 [Cl \bullet] [PVC]$$

Concentration of Alkyl Radicals ($R' \bullet$):

$$\frac{d[R' \bullet]}{dt} = k_6 [PVC] - k_7 [R' \bullet]$$

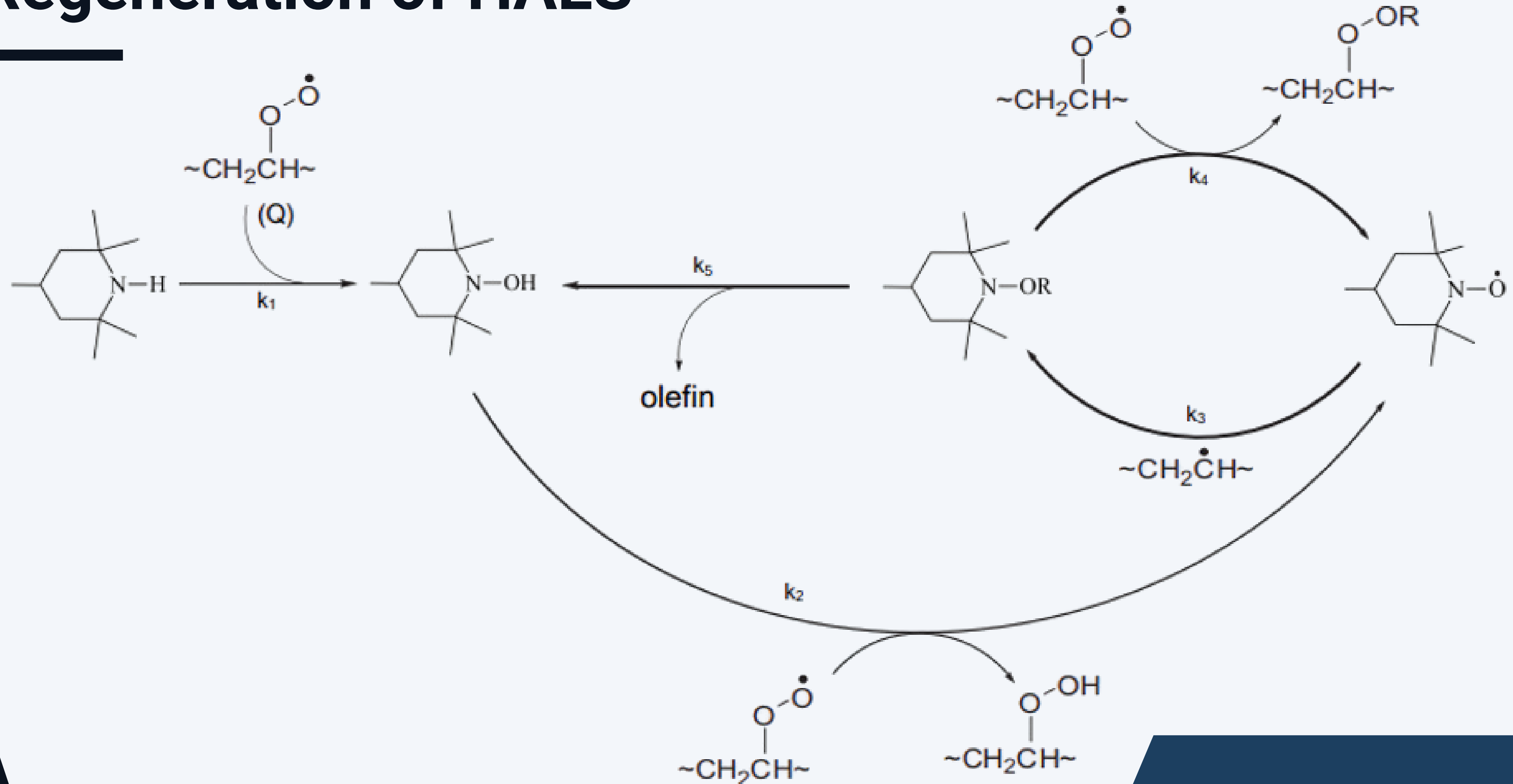
Concentration of Peroxy Radicals ($R'OO \bullet$):

$$\frac{d[R'OO \bullet]}{dt} = k_7 [R' \bullet] - k_8 [R'OO \bullet] [PVC]$$

Concentration of Peroxy Radicals ($Cl \bullet$):

$$\frac{d[Cl \bullet]}{dt} = k_6 [PVC] + k_8 [PVC] [R'OO \bullet] - k_{10} [Cl \bullet]^2$$

Radical Scavenging and Regeneration of HALS



Rate equations with stabilizers,

Concentration of PVC:

$$\frac{d[PVC]}{dt} = -k_6 [PVC] - k_8 [PVC] [R'OO \bullet] - k_9 [Cl \bullet] [PVC]$$

Concentration of Peroxy Radicals (**Cl•**):

$$\frac{d[Cl \bullet]}{dt} = k_6 [PVC] - k_8 [PVC] [R'OO \bullet]$$

Concentration of N-Hydroxy HALS (**R-N-OH**):

$$\frac{d[R - N - OH]}{dt} = k_1 [R'OO \bullet] [R - N - H] + k_5 [R - N(O - R')] - k_2 [R - N - OH] [R'OO \bullet]$$

Concentration of Alkoxyamine (**R-N(O-R')**)

$$\frac{d[R - N(O - R')]}{dt} = k_3 [R - NO \bullet] [R' \bullet] - k_5 [R - N(O - R')] - k_4 [R - N(O - R')] [R'OO \bullet]$$

Concentration of Nitroxyl Radical (**R-NO•**):

$$\frac{d[R - NO \bullet]}{dt} = k_4 [R - N(O - R')] [R'OO \bullet] + k_2 [R'OO \bullet] [R - N - OH] - k_3 [R' \bullet] [R - NO \bullet]$$

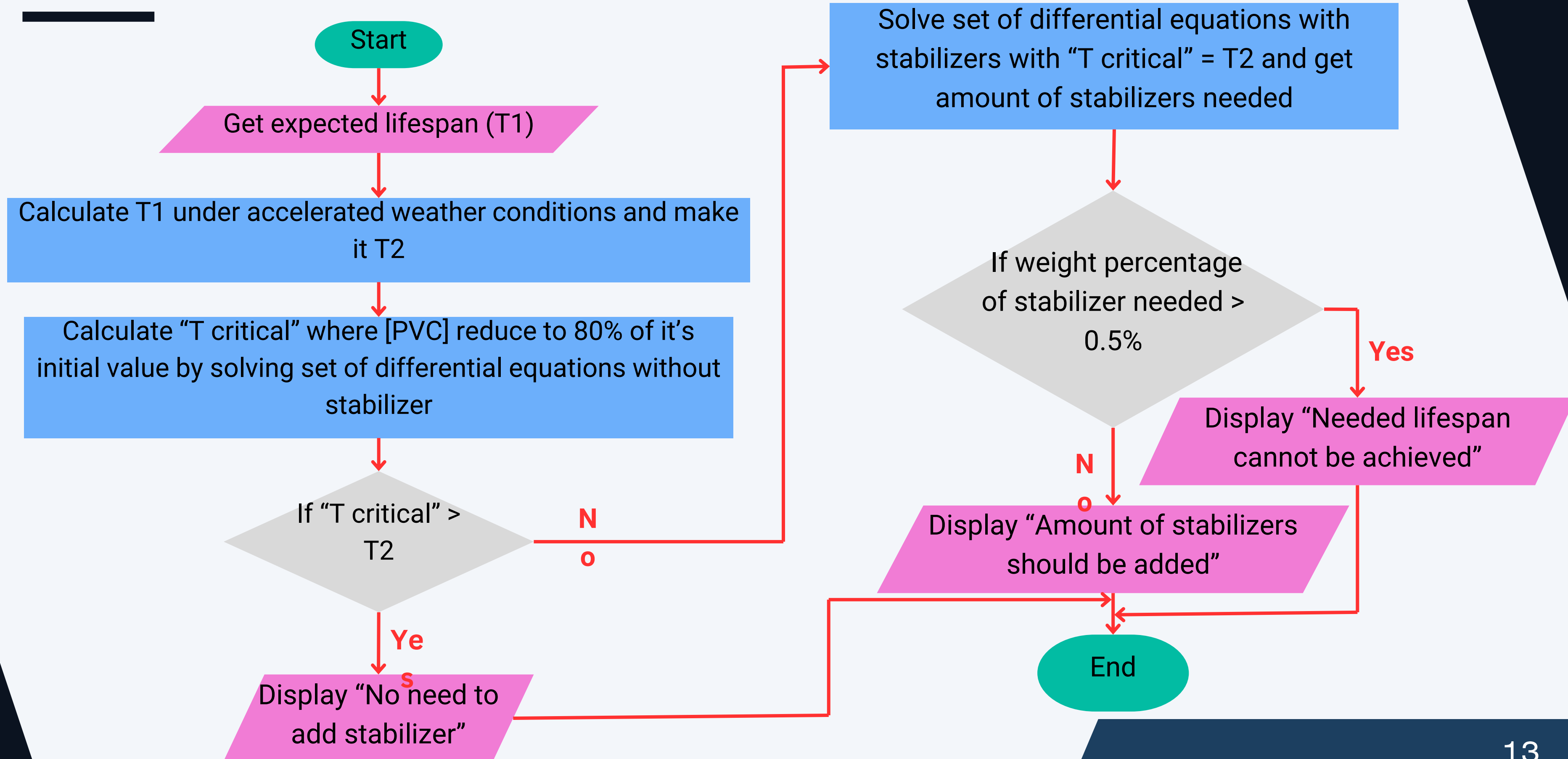
Concentration of Alkyl Radicals (**R'•**):

$$\frac{d[R' \bullet]}{dt} = k_6 [PVC] - k_7 [R' \bullet] - k_3 [R' \bullet] [R - NO \bullet]$$

Concentration of Peroxy Radicals (**R'OO•**):

$$\begin{aligned} \frac{d[R'OO \bullet]}{dt} = & k_7 [R' \bullet] - k_8 [R'OO \bullet] [PVC] - k_1 [R'OO \bullet] [R - N - H] - k_2 [R'OO \bullet] [R - N - OH] \\ & - k_4 [R - N(O - R')] [R'OO \bullet] \end{aligned}$$

Predictive Algorithm



Acceleration Factor

$$t_{env} = t_{lab} \times AF$$

Depend on

Intensity

$$AF_I = \frac{I_{lab}}{I_{env}}$$

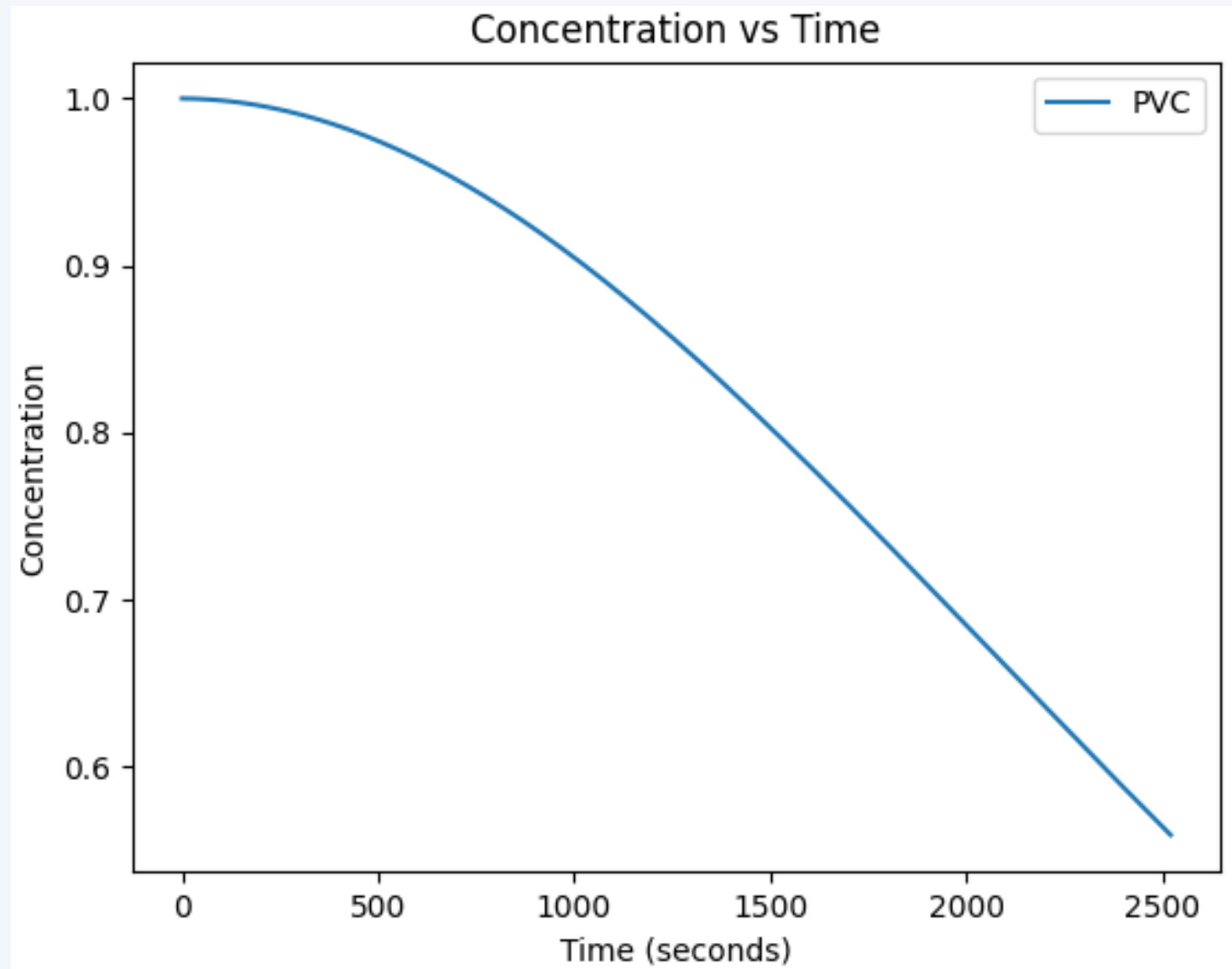
$$AF = AF_T \times AF_I$$

Depend on Temperature

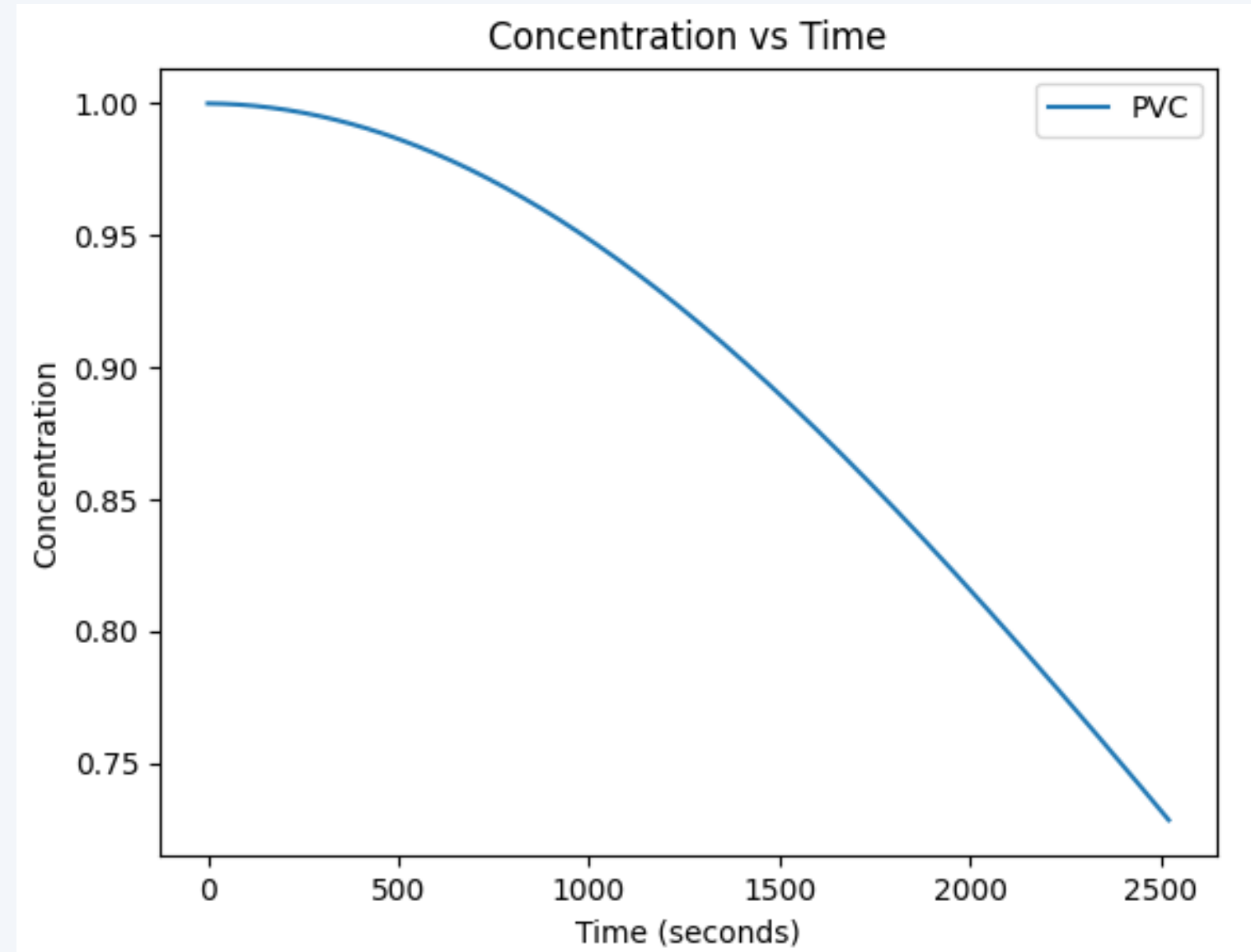
$$AF_T = \exp \left(\frac{E_a}{R} \left(\frac{1}{T_{env}} - \frac{1}{T_{lab}} \right) \right)$$

- T_{lab} : Laboratory temperature
- I_{lab} : Laboratory light intensity
- t_{lab} : Laboratory exposure time
- T_{env} : Environmental temperature (actual usage condition)
- I_{env} : Environmental light intensity

Validating Results

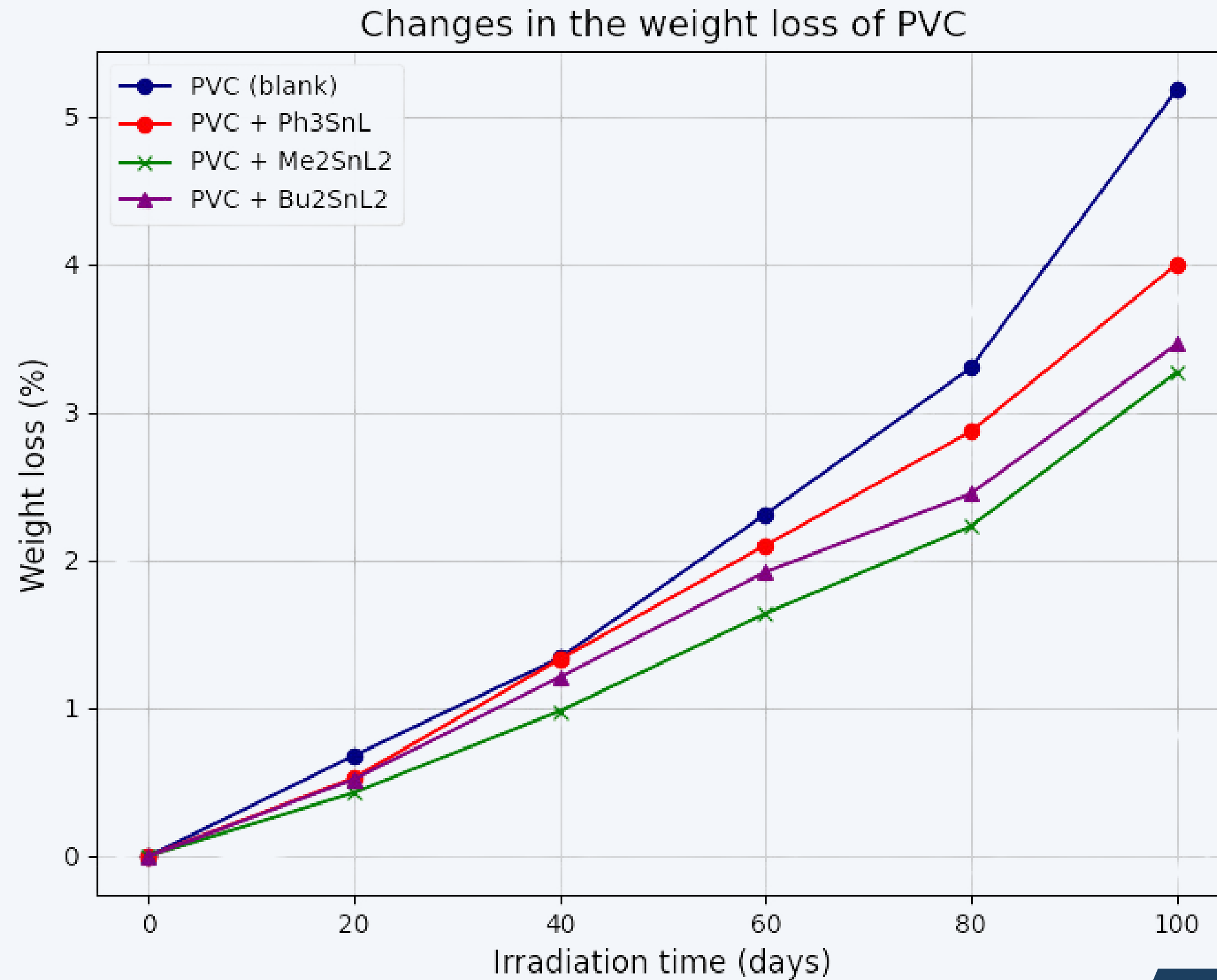


PVC concentration when stabilizer content is zero



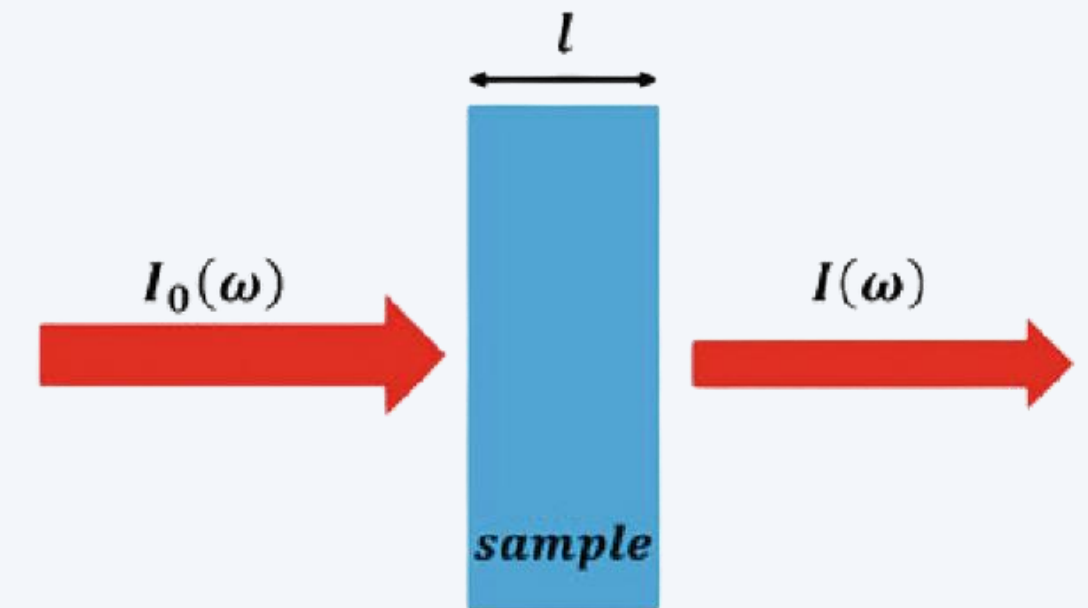
PVC concentration when stabilizer content is 0.5% wt

Experimental Data



Proposed Enhancements

- Combination with other multiple Stabilizers
- Application of Beer-Lambert Law
- UV Intensity, Time, and Temperature Dependence
- Surface Coatings
- Advanced Characterization Techniques
 - real time spectroscopic monitoring





Conclusion

- Kinetics model was developed as expected.
- Further modifications can be done.



Thank You!

Q & A

